

✉ radityapradana20@gmail.com | ✉ mohammad.raditia03@ui.ac.id | [LinkedIn](#) | [Personal Website](#) | [GitHub](#)
[ORCID 0009-0005-8836-6801](#) | [Google Scholar](#) | [ResearchGate](#) | [Scopus](#)

RESEARCH INTERESTS

Street-view imagery (SVI) • Urban data science • Computer vision • Semantic segmentation • Geospatial analysis • Urban green spaces • urban planning

EDUCATION

M.Sc. in Geography Sciences (Fast-Track) Universitas Indonesia 2024 – 2025
GPA: **4.00/4.00** | Summa Cum Laude, Faculty Level | Ranked 1st in Cohort
Thesis: *Spatial Modeling of Visual Element Proportion Diversity in Urban Spaces Based on Street-view and Spatial Indicators (Study Case: Kebayoran Baru & Tanah Abang)*

B.Sc. in Geography (Fast-Track) Universitas Indonesia 2020 – 2024
GPA: **3.85/4.00** | Cum Laude, Department Level | Ranked 1st in Cohort
Thesis: *Protection Level for Fire Prone Areas in East Jakarta City*

EXPERIENCE

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

Full-time Researcher & Assistant Editor, Smart City Journal Oct. 2024 – Present

- Geospatial research using SVI and generative AI for urban environment analysis.
- Manages [Smart City Journal](#) editorial workflow (manuscript handling, reviewer coordination).

Part-time Research Assistant Jun. – Oct. 2024

- GIS and remote sensing analysis using 3D data and digital building audits.
- Project-in-Charge for a government-collaborated urban research project.

Department of Geography, Universitas Indonesia

Researcher May 2025 – Present

- Co-PI on a university-funded grant (IDR 100 million) on climate resilience in urban parks via SVI and place attachment analysis.
- Mentors 3 undergraduate students across full research cycle.

Student Researcher Jun. 2023 – Oct. 2024

- Fieldwork on urban fire management in Jakarta; produced 1 publication and 2 international conference presentations.

Center for Social and Environmental Management Research (PPSML) — Project-Based, Universitas Indonesia

Cartographer — Strategic Environmental Assessment (SEA), West Bangka Regency Mar. – Dec. 2025

- Produced 50+ thematic and base maps at multiple cartographic scales to support the SEA document for West Bangka Regency spatial planning.

Cartographer — Strategic Environmental Assessment (SEA), Bangka Belitung Islands Province Mar. – Oct. 2024

- Produced 40+ thematic maps supporting the provincial SEA document, including environmental sensitivity and spatial constraint layers.

Center for Applied Geography Research (PPGT) — Project-Based, Universitas Indonesia

Cartographer — Detailed Spatial Plan (RDTR), Baradatu District May – Aug. 2024

- Compiled 20+ large-scale base and thematic maps at 1:5,000 for the RDTR planning document of Baradatu District, Way Kanan Regency.

Cartographer — Regional Spatial Plan (RTRW), Hulu Sungai Tengah Regency Jan. – Apr. 2023

- Produced 40+ thematic maps at 1:50,000 scale for the RTRW planning document of Hulu Sungai Tengah Regency, South Kalimantan.

TEACHING

Guest Lecturer — Master Program of Urban Planning, Universitas Indonesia

Data Management and Analysis

Apr. – May 2026

- Delivered technical lectures on database development, statistical analysis, and spatial analysis methods for graduate urban planning students.

Teaching Assistant — Department of Geography, Universitas Indonesia

GIS & Remote Sensing Practicals

Jun. – Aug. 2024

- Facilitated hands-on practicals for 30+ undergraduate students across 6 developed modules covering core GIS and remote sensing workflows.

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] **Pradana, M.R.** & Semedi, J.M. (2026). Indonesia's urban green spaces dilemma: Balancing law, ecology, and public space, highlighting Jakarta. *Human Geography*. [DOI](#)
- [2] **Pradana, M.R.**, Dimiyati, M. & Gamal, A. (2025). Harmonizing street-view semantics and spatial predictors for dominant urban visual composition modelling. *Computational Urban Science*, 5(5). [DOI](#)
- [3] **Pradana, M.R.**, Wibowo, A. & Semedi, J.M. (2025). Multi-Perspective Evaluation of Urban Green Views: Spatial and Street-View Data Integration in Sudirman Central Business District, Indonesia. *Geomatics and Environmental Engineering*, 19(3), 91–114. [DOI](#)
- [4] **Pradana, M.R.**, Dimiyati, M. & Semedi, J.M. (2025). Unveiling greenery and visual comfort: Integrating Green View Index and image segmentation in panoramic rural landscapes. *GeoScape*, 19(2), 143–159. [DOI](#)
- [5] **Pradana, M.R.** & Dimiyati, M. (2024). Tracking urban sprawl: A systematic review and bibliometric analysis of spatio-temporal patterns using remote sensing and GIS. *European Journal of Geography*, 15(3), 190–203. [DOI](#)
- [6] Gamal, A., **Pradana, M.R.**, Wibisono, B.H., Navitas, P. & Aryal, J. (2025). Thematic fragmentation and convergence in urban flood simulation research: A 45-year bibliometric mapping. *Urban Science*, 9(7), 280. [DOI](#)
- [7] Semedi, J.M., **Pradana, M.R.**, Putera, D.A. & Rahatiningtyas, N.S. (2024). The impedance of East Jakarta road network to estimate fire station service areas. *Forum Geografi*, 38(2), 280–290. [DOI](#)

Conference & Short Communications

- [1] **Pradana, M.R.** & Semedi, J.M. (2025). Effortless coastal monitoring: Unsupervised detection of shoreline alterations due to tin mining in Bangka Belitung. *IOP Conference Series: EES*, 1462(1). [DOI](#)
- [2] Ash-Shidiq, I.P. et al. [**Pradana, M.R.** incl.] (2024). Biomass stock estimation using Landsat 8 imagery in Bukit Tigapuluh National Park, Riau. *IOP Conference Series: EES*, 1291(1). [DOI](#)

Under Review & In Development

- In Review** *Your Academic Career Depends on Your Co-Authors: When Single-Author Papers Do Not Count in Indonesia's Academic Promotion System.* **Journal of Academic Ethics.**
- In Review** *Reframing urban "sustainable" streetscapes: Evidence from cross-city transformation and convergence.* **City and Environment Interactions.**
- In Review** *Interpreting 'sustainable' urban streetscapes with generative AI: Context-rich vs. generic prompting.* **Urban Informatics.**

OPEN-SOURCE SOFTWARE & DATA

Cooling Cities Project Page — Research dissemination website

2025  [Page](#)

Prompting Sustainability Project Page — Research dissemination website

2025  [Page](#)

GSV Panorama Downloader — Notebook repo for batch downloading Google Street View panoramas

2025  [GitHub](#)

GSV Downloader by Links — Notebook repo for downloading Street View imagery via direct links

2025  [GitHub](#)

Open-Access Datasets

genai_sustainablestreetscape (v2) — Cross-city corpus of 4,186 paired images (raw + GPT-5-generated "more sustainable" counterparts) from Jakarta (567), Melbourne (832), and Singapore (694). All scenes standardised to 640×640 px; includes source coordinates and generation meta-data.

2026  [Hugging Face](#)

Sustainable Streetscape Prompt-Specificity Dataset (v1) — 100 Jakarta GSV scenes × 3 treatments (raw, context-rich prompt, non-context prompt via GPT-4o), each paired with semantic segmentation masks. Enables comparative study of how prompt specificity reshapes AI interpretation of urban sustainability. 2026  [Zenodo](#)

RESEARCH & GOVERNMENT PROJECTS

Research Grants

Prompting Sustainability: Generative AI and Urban Sustainable Streetscapes Funded by Universitas Indonesia	Oct. 2025 – Present
Cooling Cities, Warming Communities: Climate Resilience in Urban Parks IDR 100 million, 2-year grant — Universitas Indonesia	May 2025 – Present
Spatial Modeling of Visual Element Proportion Diversity [Master's Thesis] INSPIRASI Grant — SMART CITY UI	Jun. 2024 – Jun. 2025
Collaborative Mapping & Mitigation for a Resilient Capital City with Generative AI PRIME Engineering Grant (seed; potential IDR 1B follow-up)	Jun. 2024 – Dec. 2025
Level of Protection of Fire-Prone Areas in East Jakarta City Funded by Faculty of Mathematics and Natural Sciences, Universitas Indonesia	Aug. 2023 – Jan. 2024

Government Projects

Innovative Government Awards 2025, Ministry of Home Affairs (<i>PIC & data scientist; 12,000+ innovations assessed</i>)	Sep. – Nov. 2025
Strategic Environmental Assessment (SEA) — West Bangka Regency	Mar. – Dec. 2025
Innovative Government Awards 2024, Ministry of Home Affairs (<i>PIC & data scientist; 8,000+ innovations assessed</i>)	Sep. – Oct. 2024
Strategic Environmental Assessment (SEA) — Bangka Belitung Islands Province	Mar. – Oct. 2024

ACADEMIC PEER REVIEW

<i>Journal of Mountain Science</i> , Springer Nature	May. 2026
<i>Scientific Reports</i> , Nature Publishing Group	Jan. – Feb. 2026
<i>Frontiers in Marine Sciences</i> , Frontiers	Jan. – Feb. 2026
<i>Urban Forestry & Urban Greening</i> , Elsevier	May – Aug. 2025
<i>Transactions in GIS</i> , Wiley	Jan. 2025

CONFERENCE PRESENTATIONS

08/2024	ICoAGPG — <i>Effortless shoreline detection in tin-mining areas.</i> [Best Presenter]
11/2023	AICGEE — <i>The Impedance of East Jakarta Road Network to Estimate Fire Station Service Areas.</i>
11/2022	ICoSAG — <i>Biomass Stock Estimation Using Landsat 8 Imagery in Bukit Tigapuluh National Park.</i>

HONORS & AWARDS

2025	Ranked 1st in Cohort, M.Sc. Geography — Summa Cum Laude (Faculty Level)
2024	Ranked 1st in Cohort, B.Sc. Geography — Cum Laude (Department Level)
2024	Best Presenter , ICoAGPG 2024
2022	2nd Place , National Economics Paper Competition (IMIE FEB UKSW) — spatial-economic policy proposal aligned with SDGs

TECHNICAL SKILLS

Well-versed in GIS both through GUI-based platforms (QGIS, ArcGIS, Google Earth Engine) and code-based workflows in **Python**; familiar with R. Specialises in **street-view imagery (SVI)** analysis — from data acquisition (Google Street View API, historical panorama fetching) through panorama processing, to semantic segmentation using state-of-the-art models (Mask2Former, OneFormer) trained on ADE20K and Cityscapes, enabling pixel-level quantification of urban visual elements such as greenery, sky, and built surfaces. Applies derived indices (Green View Index, visual comfort metrics) and dimensionality reduction (UMAP, t-SNE) to characterise urban environments at scale. Experienced in **generative AI** for urban research, including model fine-tuning, prompt engineering, and AI-assisted image interpretation and generation. Has led and contributed to multiple research projects across the full cycle — proposal development, budget planning, data collection and analysis, through to formal reporting and dissemination. Conducts **systematic**

literature reviews with bibliometric mapping, thematic synthesis, and structured assessment of large bodies of scholarly literature.